

Is low folate a risk factor for depression? A meta-analysis and exploration of heterogeneity

Low folate has been causatively linked to depression, but research is contradictory. An association may arise due to chance, bias, confounding or reverse causality. A systematic review of observational studies which examined the association between depression and folate was conducted. 11 relevant studies (15 315 participants; three case-control studies, seven population surveys and one cohort study) examining the risk of depression in the presence of low folate were found. Pooling showed a significant relationship between folate status and depression (odds ratio (OR)pooled unadjusted = 1.55; 95% CI 1.26 to 1.91). This relationship remained after adjustment for potential confounding (OR)pooled adjusted = 1.42; 95% CI 1.10 to 1.83). Folate levels were also lower in depression. There is accumulating evidence that low folate status is associated with depression. Much of this evidence comes from case-control and cross-sectional studies. Cohort studies and definitive randomised-controlled trials to test the therapeutic benefit of folate are required to confirm or refute a causal relationship.